



Influence of zirconium on the structure of pristine and leached soda-lime borosilicate glasses: Towards a quantitative approach by ^{17}O MQMAS NMR

Frédéric Angeli^{a,*}, Thibault Charpentier^b, Marina Gaillard^b, Patrick Jollivet^a

^a Laboratoire d'Étude du Comportement à Long Terme, DEN/DTCD/SECM, CEA Valrhô, 30207 Bagnols-sur-Cèze cedex, France

^b Laboratoire de Structure et Dynamique par Résonance Magnétique, DSM/DRECAM/SCM, CEA Saclay, 91191 Gif-sur-Yvette cedex, France

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ABSTRACT

^{17}O MQMAS NMR was used to characterize the influence of zirconium on the structural organization of soda-lime borosilicate glasses. A new method of quantitative analysis of the ^{17}O MQMAS spectra is presented, by a direct fit of the two-dimensional MQMAS spectrum which provides the resolution of all the structural groups in glasses containing up to five oxides. Additional data were also obtained from the quantitative deconvolution of the ^{11}B MAS NMR spectra, with the help of the direct fit of MQMAS data as well. Excess of non-bridging oxygen is clearly identified in these glasses. Sixfolded zirconium is preferentially compensated rather than the tetrahedral boron and calcium only partially compensate the tetrahedral boron. Alteration gels arising from glass leaching were probed by oxygen-17 supplied by the alteration solution. Most of the zirconium is inserted in the silicate network forming Si–O–Zr bonds with the same configuration in the glass and in the gel. During leaching, calcium clearly remains in the alteration gel, either near non-bridging oxygen or as a zirconium charge compensator. This quantitative approach applied to ^{17}O MQMAS spectra demonstrates its potential for investigating the structure of increasingly complex glass and gel compositions.

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1. Introduction

The alteration layer (commonly known as the ‘gel’) that forms during the alteration of borosilicate glasses by water account for a drop in the glass alteration rate and can develop retention properties with respect to some vitreous matrix constituent elements [1]. These properties could be significantly influenced by the gel structure itself. During leaching, rearrangements among structural groups are related to the pristine glass chemical composition and could also depend in part on the initial glass structure [2]. Structural characterization of the gel according to the glass composition can thus be useful for a better understanding of the alteration mechanisms. These issues affect many areas for which the reactivity of the glass must be known: conditioning glass, archaeological glass, or radioactive waste containment glass repositories. The particular role of zirconium in the alteration mechanisms of five-oxide glasses has been recently highlighted, demonstrating the strong influence of this element on the morphology and properties of

the alteration layer [3]. In the present work, this issue is addressed by a structural characterization of glasses and their alteration gels, as provided by NMR spectroscopy.

As being bonded to both network formers and modifiers, the oxygen atom is an ideal probe of the structure of oxide glasses, with the requirement of oxygen-17 isotopic enrichment. However the standard sample magic-angle spinning (MAS) technique cannot resolve the different oxygen sites when the glass contains several oxides. Thus the use of two-dimensional experiments, such as the MQMAS (Multiple Quantum Magic Angle Spinning) experiment [4] is required. MQMAS has been successfully applied to glasses containing up to four oxides, mainly borates, borosilicates, aluminosilicates and aluminoborosilicate containing alkalis or alkaline earths, and in which bridging and non-bridging oxygen groups were identified [5–11]. Its strong potential for characterizing the structural groups forming the gel has also been pointed out. The reconstruction of an aluminosilicate gel with the same structural configuration as the pristine glass but without non-bridging oxygens (NBO) was observed during leaching [12].

In this work, it is first pointed out that resolving all the structural groups in a glass containing up to five oxides can be achieved with the help of MQMAS. The combined use of ^{11}B and ^{17}O NMR

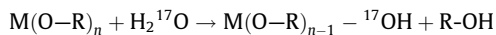
* Corresponding author.

E-mail addresses: frederic.angeli@cea.fr (F. Angeli), Thibault.charpentier@cea.fr (T. Charpentier).

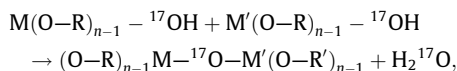
data, and structural models may be used to inform about the short-range atomic organization around structural units requiring charge compensation such as fourfolded boron and sixfolded zirconium. Two glass compositions were investigated: the first is comprising four oxides ($70\text{SiO}_2\text{--}18\text{B}_2\text{O}_3\text{--}10\text{Na}_2\text{O--}2\text{CaO}$) and the second five oxides ($67\text{SiO}_2\text{--}16\text{B}_2\text{O}_3\text{--}10\text{Na}_2\text{O--}3\text{CaO--}4\text{ZrO}_2$). In the second part of this study, glasses of similar compositions but with various Na content were altered in oxygen-17 enriched water, thus providing a selective characterization of the exchange sites between the solution and the glass. Such information on the gel structure may then be compared with the pristine glass structure. The distribution of alkalis and alkaline earths in the glass and gel structure is another item of interest in this type of composition. Examining changes between network-modifying and charge-compensating positions can lead to a better understanding of the leaching mechanisms.

2. Experimental conditions

Fabrication of glasses enriched in oxygen-17. The glasses chemical composition is indicated in Table 1. The materials were synthesized by the sol-gel process, involving hydrolysis (70% enriched water) and polycondensation from metal alkoxides. All the elements were used under alkoxide form ($\text{Si}(\text{OC}_2\text{H}_5)_4$, $\text{B}(\text{OC}_4\text{H}_9)_3$, $\text{Na}(\text{OC}_2\text{H}_5)$, $\text{Zr}(\text{OC}_3\text{H}_7)_4$) and mixed together in anhydrous ethanol. The metal precursors used for glass fabrication were enriched according to the following reactions:



and



where M and M' are cations; R and R' are organic groups.

Table 1
Molar composition of glasses enriched in oxygen-17

	SiO_2 (mol%)	B_2O_3 (mol%)	Na_2O (mol%)	CaO (mol%)	ZrO_2 (mol%)
^{17}O -Si glass	70.0	17.8	9.9	2.3	–
^{17}O -Si-Zr glass	67.2	15.5	10.4	3.3	3.6

Table 2
Molar composition of pristine glasses and gels

	Composition (mol%)						Retention factor (%)			
	SiO ₂	B ₂ O ₃	Na ₂ O	CaO	ZrO ₂	Gel %	Si	Zr	Ca	Na
Series 1										
O–Si glass 1	61	17	18	4	–					
O–Si–Zr glass 1	56	18	18	4	4					
Si gel 1	92	0	0	8	–		71		97	0
Si–Zr gel 1	87	0	2	5	6	5	91	100	79	5
Alteration time							176 days			
Series 2										
O–Si glass 2	64	18	14	4	–					
O–Si–Zr glass 2	60	18	14	4	4					
Si gel 2	93	0	0	7	–	3	86	–	98	0
Si–Zr gel 2	84	0	4	6	6	2	99	100	99	21
Alteration time							652 days			
Series 3										
O–Si glass 3	70	18	10	2	–					
O–Si–Zr glass 3	67	16	10	3	4					
Si gel 3	97	0	0	3	–	10	97	–	96	0
Si–Zr gel 3	88	0	2	5	5	2	93	100	98	14
Alteration time							526 days			

The gel percentage is indicated in the column headed 'Gel %'. The right-hand part of the table indicates the element retention factors in the gel (compared with boron, inferred from solution analysis after different leaching time in pure water).

The quantity of H_2^{17}O added was the exact stoichiometry according to the first reaction. Glass (500 mg) was prepared using 600 mg of enriched water. The advantage of this method is that it can enrich all the glass precursors in oxygen-17. The resulting precursors were melted for 15 min on platinum foil at 1220 °C. The glass was analyzed by alkaline fusion ($\text{NaOH} + \text{KNO}_3$ and $\text{Li}_2\text{B}_4\text{O}_7 \cdot 5\text{H}_2\text{O}$) of powder samples (~50 mg) then dissolved in HNO_3 for ICP-AES analysis. A paramagnetic element (Gd_2O_3) was added to the glass in very small quantities (0.2 wt%) to diminish the relaxation time of the ^{17}O nucleus during NMR spectrum acquisition.

Fabrication of non-enriched glasses. Glasses were prepared from powder mixtures of SiO_2 , H_3BO_3 , Na_2CO_3 , CaO , and ZrO_2 melted in a platinum crucible and refined for 3 hours. The process temperatures ranged from 1300 °C to 1400 °C. The analyzed glass compositions are indicated in Table 2.

Leaching. Glasses without oxygen-17 enrichment were leached in contact with water enriched in oxygen-17 (with an enrichment between 40 and 70%). The glass specimens were ground in tungsten carbide bowls then screened to recover the particle size fraction between 125 and 250 μm . The powder was washed in acetone to eliminate the fine glass particles. The specific surface area of this particle size fraction measured by krypton adsorption using the BET method was about 380 $\text{cm}^2 \text{g}^{-1}$. Glass specimens were then leached in static mode, unstirred, at 90 °C for 176 days until 652 days with a glass-surface-area-to-solution-volume ratio (SA/V) of 120 cm^{-1} . The quantity of altered glass was calculated from the boron release, as boron is a good glass alteration tracer [13].

MAS and MQMAS NMR. All spectra were collected on a Bruker Avance II 500WB spectrometer (magnetic field 11.74 T), using a Bruker 4 mm (o.d.) CPMAS probe at a spinning frequency ranging from 12.5 kHz to 14 kHz.

For ^{11}B ($I = 3/2$, $\nu_0 = 160.14 \text{ MHz}$), direct acquisition spectra were obtained with repetition times of 2 s using a single short pulse (1 μs) to obtain quantitative spectra (nuclei with spins higher than $1/2$ are subjected to a quadrupolar interaction that interferes with the radiofrequency field during the excitation pulse, resulting in a non-quantitative spectrum except with very short pulses corresponding to linear conditions [14]). Chemical shifts are referenced to an external sample of 1 M boric acid solution

(19.6 ppm). Because of the large difference in the quadrupolar coupling constant C_Q between the three and fourfold boron species (see text), ^{11}B Triple Quantum MQMAS experiments were specifically optimized for the observation of each species. For fourfold species, spectra were acquired using the Z-filter pulse sequence [15]. The lengths of the triple quantum excitation and reconversion pulse were 5.5 μs and 2.1 μs (RF field of 100 kHz) and the soft 90° pulse was set to 5 μs (RF field 25 kHz). 128 t_1 increments of 80 μs were collected with a delay of 2 s. For threefold boron species, a shifted echo pulse sequence [16] was performed. The lengths of the triple quantum excitation and reconversion pulse were 5.5 μs and 20 μs (RIACT pulse [17], RF field of 100 kHz), the soft 180° pulse was set to 10 μs (RF field 25 kHz) and an echo delay of 4.8 ms was used. 64 t_1 increments of 10 μs were collected with a delay of 1 s.

For ^{17}O ($I = 5/2$, $\nu_0 = 67.67$ MHz), the MQMAS spectra were acquired at a sample spinning frequency of 14 kHz, with 71.5 μs increment in the first dimension (synchronized with the rotation, 64 points) and a 1 s repetition time (4500 FIDs per t_1 value). The Z-filter pulse sequence [15] was used with first and second pulse durations of 7 μs and 2.6 μs , respectively (RF field 60 kHz), and a third 90° soft pulse selective on the central transition of 5.5 μs (RF field of 15 kHz) [18]. ^{17}O MAS spectra were acquired using a Hahn Echo pulse sequence, with soft 90° and 180° pulses (selective on the central transition) and a echo delay of one rotation period so as to minimize baseline distortion.

All spectra were analyzed (processing and simulation) with an homebuilt software, using a theoretical approach described in [19]. In the present work, the dependence of the efficiency of the MQMAS pulse sequence upon the quadrupolar interaction was taken into account in order to devise a quantitative analysis of the MQMAS data. Some details can be found in [20], as well as the principles of modeling MQMAS spectra of amorphous systems. More details on the simulation program devised for the present study will be given elsewhere.

3. Results and discussion

3.1. Towards quantitative analysis of ^{11}B and ^{17}O MQMAS NMR data

For a quantitative analysis of NMR data of amorphous materials such as glasses, the main difficulty often rely in the consideration of the broadening resulting from the structural disorder, leading to distributed NMR parameters, in order to properly fit the NMR lines (MAS or MQMAS). For ^{23}Na , ^{27}Al and ^{43}Ca NMR, it has been recently demonstrated that the Gaussian isotropic model (GIM) [21] could be used for the distribution of the quadrupolar interactions parameters (arising from a large distribution of the local Electric Field Gradient, EFG), yielding highly accurate fitting of the experimental data [21,22].

Here, such a new methodology is introduced to process both ^{11}B and ^{17}O MQMAS spectra so as to extract NMR parameters of each site and quantify its respective population. The method consist in a direct fitting of the MQMAS spectra, taking into account the distribution of each NMR parameters, the effects of the pulse sequence and of the sample rotation on the intensity and the lineshape of the NMR spectra [20]. In contrast to ^{23}Na , ^{27}Al or ^{43}Ca environments, ^{11}B and ^{17}O are characterized by a much narrower distribution of quadrupolar interaction values, so that a product of three Gaussians has been considered, one for each NMR parameter i.e. the quadrupolar coupling constant C_Q , the quadrupolar asymmetry parameter η_Q and the isotropic chemical shift δ_{iso}

$$P(\delta_{\text{iso}}, C_Q, \eta_Q) = G(\delta_{\text{iso}}) \times G(C_Q) \times G(\eta_Q)$$

with

$$G(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{(x - \bar{x})^2}{2\sigma_x^2}\right).$$

No correlation between the NMR parameters was taken into account; this should be presented elsewhere, as well as details about the simulation algorithms. For the quadrupolar asymmetry parameter η_Q , the distribution is truncated for values in the range [0,1]. The present procedure yields for each sites, mean NMR parameters as well as their respective standard deviation, denoted σ_x ($x = C_Q, \eta_Q$ and δ_{iso}).

A binary glass ($24\text{Na}_2\text{O}-76\text{SiO}_2$) was used to validate the quantification procedure of ^{17}O 3QMAS spectra. The experimental spectrum and simulated ^{17}O 3QMAS are displayed in Fig. 1(a) and (b), respectively. Fig. 1(c) gives the isotropic projections of each spectrum (experimental (top) and simulated (middle)) as well as the direct fit of the experimental isotropic dimension showing each component (bottom, using Gaussian lineshapes). The latter is shown because the direct fit of the isotropic projection is the only procedure that have been applied so far (see [9] for example) for quantifying ^{17}O 3QMAS and is given here for purpose of comparison. Its major drawback relies in its limitation to simple glass compositions so that the isotropic projection exhibits a small overlapping between the different lines. For complex compositions, such as those investigated in the present work, it cannot be applied as a result of a strong overlapping of all lines in the isotropic dimension, resulting in a ambiguous quantification. The MQMAS excitation profile versus the quadrupolar coupling, as constant displayed in Fig. 1(d) (for the used 3QMAS sequence and only displayed for a quadrupolar asymmetry parameter $\eta = 0.5$), can be used to correct each line intensities so as to provide correct site populations from the direct fit of the isotropic projection. As shown in Fig. 1(b) and (c), the simulated spectra are in close agreement with the experimental data. To the best of our knowledge, this is the first attempt to a direct quantitative simulation of the ^{17}O MQMAS spectra.

Tables 3 and 4 give the population of each site of $24\text{Na}_2\text{O}-76\text{SiO}_2$ glass using the two different methods; from fitting the isotropic projection and correcting intensities with the efficiency curve (Fig. 1(d) and Table 3); from the direct simulation of the MQMAS spectrum as described before (Table 4). A very good agreement is found between both approaches and consistent values with theoretical ones (72.7 Si–O–Si and 27.3 Si–O–Na) are obtained.

3.2. Glass structure

Two soda-lime borosilicate glasses enriched in oxygen-17, without zirconium (designated as $^{17}\text{O}-\text{Si}$) and with zirconium (designated as $^{17}\text{O}-\text{Si}-\text{Zr}$) (Table 1) have been characterized by ^{11}B and ^{17}O NMR.

3.2.1. ^{11}B MAS and MQMAS NMR

^{11}B MAS NMR is now commonly used to quantify the proportion of three-coordinate (BO_3) and four-coordinate ($[\text{BO}_4]^-$) species. For the quantitative analysis of ^{11}B MAS NMR spectra, especially for its deconvolution into BO_3 and BO_4 lines, 3QMAS experiments optimized for the observation of tri-coordinated and four-coordinate boron sites, respectively (see experimental section), were recorded for the $^{17}\text{O}-\text{Si}$ glass. They were separately fitted using procedure described in the preceding section. It was found that at least two sites were necessary to correctly simulate the MQMAS spectrum of the BO_3 sites, as displayed in Fig. 2 (simulation parameters are given in Table 5). According to previous work [5], these two sites can be attributed to ring and non-ring three-folded boron sites. The obtained parameters were used as initial values for fitting the MAS spectra using the same kind of distribution. It should be

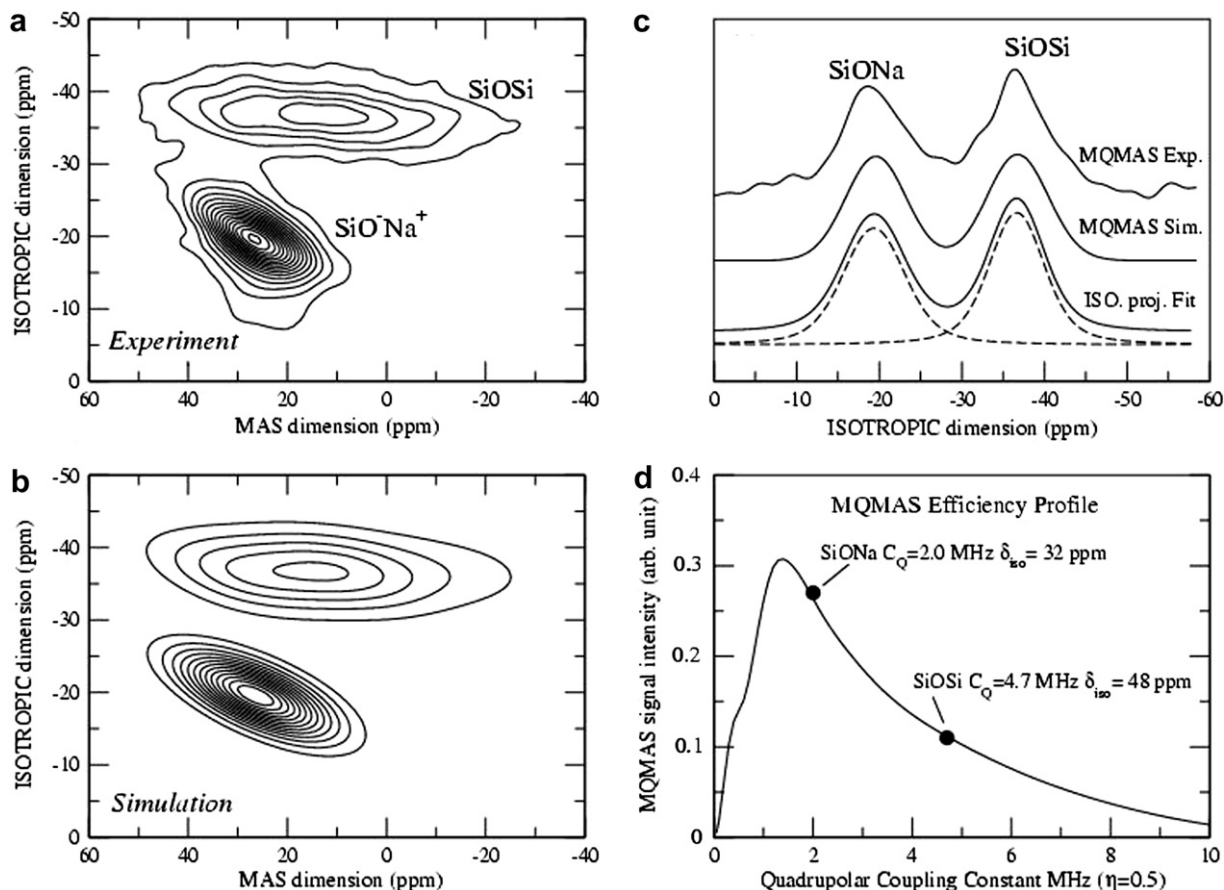


Fig. 1. (a) Experimental and simulated (b) ^{17}O 3QMAS spectrum of $24\text{Na}_2\text{O}-76\text{SiO}_2$ glass. (c) Isotropic projections of the experimental (Exp.) and simulated (Sim.) spectra, and fit of the experimental isotropic projection showing each component (dashed lines). (d) Excitation (or efficiency) profile versus quadrupolar coupling constant C_Q for the 3QMAS sequence ($I = 5/2$) with quadrupolar asymmetry parameter $\eta_Q = 0.5$.

Table 3
 ^{17}O NMR parameters obtained from the direct simulation of the MQMAS spectra ($24\text{Na}_2\text{O}-76\text{SiO}_2$ glass)

Site	C_Q (MHz)	σ_{C_Q} (MHz)	η_Q	σ_{η_Q} (MHz)	δ_{iso} (ppm)	$\sigma_{\delta_{\text{iso}}}$ (ppm)	Proportion (experimental)	Proportion (theoretical)
Si–O–Si	4.73	0.26	0.5	0.4	48.3	6.2	71.7	72.7
Si–O–Na	2.03	0.21	0.6	0.1	32.2	6.6	28.3	27.3

Table 4
 ^{17}O NMR parameters obtained from simulation of the isotropic spectra and efficiency curve ($24\text{Na}_2\text{O}-76\text{SiO}_2$ glass)

Site	Proportion (experimental) ^a	Efficiency	Proportion (experimental) ^b	Proportion (theoretical)
Si–O–Si	52.5	0.11	72.3	72.7
Si–O–Na	47.5	0.26	27.7	27.3

^a As obtained from fitting the isotropic projection with Gaussian lineshapes.
^b From as obtained from fitting the isotropic projection with Gaussian lineshapes and taking into account the MQMAS efficiency.

pointed out that, in the present work, we focus on the quantification of the BO_3 and $[\text{BO}_4]^-$ unit species so that the NMR parameters were re-optimized so as to fit the MAS spectrum but only small variation (about $\sim 5\%$) were observed for each parameters. For the BO_4 line, it was also found that at least two components were necessary (Fig. 2). Additional Lorentzian broadening was also taken into account and found to improve the fit of the four-coordinated boron line on the MAS spectra. This Lorentzian broadening account for the effect of the small quantity of paramagnetic impurities incorporated in the samples in order to shorten the relaxation

times. Simulation parameters of the MAS spectra parameters are given in Table 6 (see Fig. 3).

The Dell and Bray model established for ternary borosilicate glass [23–25] enables to estimate the proportions of three-coordinate (N_3) and four-coordinate (N_4) boron according to the chemical composition assuming the existence of particular borosilicate units (diborates, reedmergnerites, etc.). The model is based on the $R = \text{Na}_2\text{O}/\text{B}_2\text{O}_3$ and $K = \text{SiO}_2/\text{B}_2\text{O}_3$ molar ratios. In borosilicates the number of N_4 increases proportionally with the total number of alkalis up to a maximum value (for $R = 1$) above which it stabilizes and then linearly diminishes as the sodium fraction increases. The degree of mixing between boron and silicon is generally greater than predicted by the model [26].

It is clearly observed in Table 6 that adding zirconium to the glass results in a significant drop in the number of four-coordinate boron atoms. In complex borosilicate glasses, X-ray absorption spectroscopy has shown that zirconium was at six fold coordinated sites, connected to silicon tetrahedra, with $[\text{ZrO}_6]^{2-}$ groups requiring compensation by two positive charges [27]. Usually, when other network formers requiring charge compensation are added, the number of N_4 becomes proportional to the number of alkalis still available after compensating the formers [28,29]. In alumin-

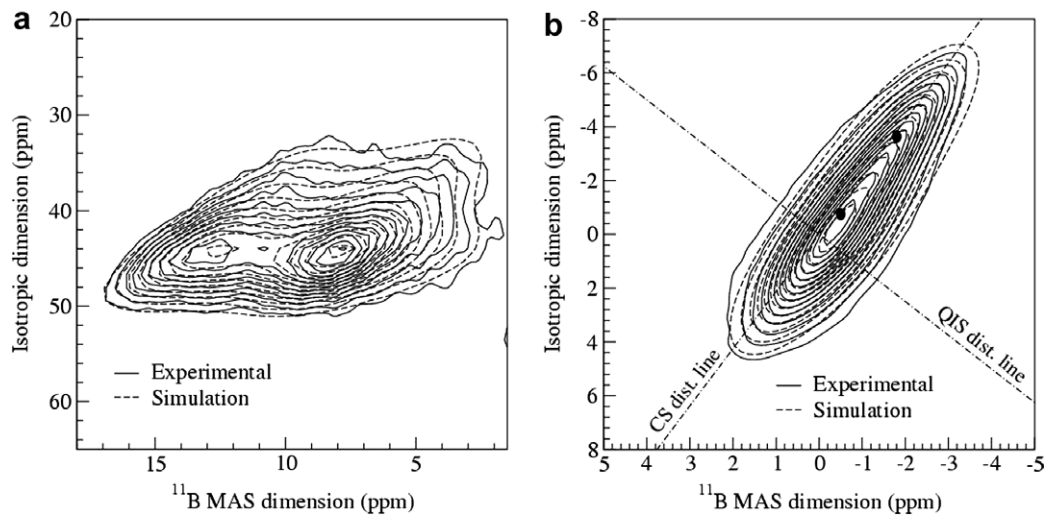


Fig. 2. Experimental (solid lines) and simulated (dashed lines) ^{11}B 3QMAS spectra of ^{17}O -Zr glass showing the region of tri-coordinated boron (a) and four-coordinated boron (b). Simulation parameters are given in Table 5.

Table 5

NMR parameters used for the simulation of the ^{11}B 3QMAS spectrum for the tri-coordinated boron

Site	δ_{iso} (ppm)	C_Q (MHz)	η	Proportion (%)
BO_3 ring	17.2 (1.4)	2.60 (0.12)	0.28 (0.1)	57
BO_3 nonring	14.9 (2.3)	2.48 (0.12)	0.27 (0.1)	43

Uncorrelated Normal distribution of C_Q , η and δ_{iso} was used (see text), standard deviation are given in parentheses.

oborosilicates, the $[\text{AlO}_4]^-$ groups are generally compensated rather than the $[\text{BO}_4]^-$ groups, in proportions that depend on the type of network-modifying cation [30]. By assuming that zirconium exhibits the same behavior as aluminum in terms of preferential compensation, experimental data are compared with a modified Dell and Bray model in Table 6 based on the following assumptions: in the first case, the model is applied by removing the cations necessary to compensate zirconium and assuming that the role of one calcium atom is equivalent to two sodium atoms; the R ratio becomes $(\text{Na}_2\text{O} + \text{CaO} - \text{ZrO}_2)/\text{B}_2\text{O}_3$. In the second case it is postulated that only sodium can compensate the charge of four-coordinate boron or zirconium, and that all the calcium forms NBOs ($R = (\text{Na}_2\text{O} - \text{ZrO}_2)/\text{B}_2\text{O}_3$). In ^{17}O -Si and ^{17}O -Si-Zr glasses, 56% (respectively 44%) of the boron should be found four-coordinated if sodium was the only charge compensator. The higher experimental values of 66% (respectively 52%) indicates that a part of the calcium must be charge-compensator, as boron charge compensation indiscriminately by either Na or Ca would result in 69%

(respectively 65%) of $[\text{BO}_4]^-$. The quantity of calcium as charge compensator was calculated to fit the experimental values for both glasses. The best agreement was found for about 50% of calcium as compensator of B and Zr (indicated in the last column of Table 6). Calcium atoms could then play a double configuration of network compensator and network modifier by forming $\text{Si}-\text{O}^{2-}-\text{Ca}^{2+}-[\text{BO}_4]^-$ groups. In this case, it means that addition of a rare-earth (CaO) in a glass in replacement of an alkali (Na_2O) should result in a decrease of tetrahedral boron. This behavior has already been experimentally observed: a significant drop in four-coordinate boron was noted in aluminoborosilicate glass if 20% of the K_2O is replaced by 20% of CaO [10].

3.2.2. ^{17}O MAS and MQMAS NMR

Complementary data can be obtained using the ^{17}O nucleus, which provides an overview of the structural groups. The results of the fitting procedure (Section 3.1) for the ^{17}O -Si glass is displayed in Fig. 4. In order to stabilize the fitting procedure, some constraints have been added such as the sharing between the BO sites of the same σ_x parameters ($x = C_Q, \eta_Q, \delta_{\text{iso}}$). The same constraints were applied to NBO sites but only for the quadrupolar parameters ($x = C_Q, \eta_Q$) as the experimental spectrum clearly shows that NBO sites differ in their distribution of isotropic chemical shift. Experimental and simulated ^{17}O 3QMAS spectra shown in Fig. 4(a) and (b), respectively, are in good agreement. Fig. 4(c) compares the MAS projection of the experimental and simulated 3QMAS spectra, also displaying the contribution of each component. Fig. 4(d) displays the distribution of NMR parameters of each site in the plane

Table 6

NMR parameters obtained for simulations of ^{11}B MAS spectra and comparison of site quantification with modified Dell and Bray model

Glass	Site	δ_{iso} (ppm)	C_Q (MHz)	η	Relative intensity	D&B Model $R = (\text{Na}_2\text{O} + x\text{CaO} - \text{ZrO}_2)/\text{B}_2\text{O}_3$		
						$x = 1$	$x = 0$	$x = 0.5$
^{17}O -Si glass	BO_3 ring	17.5(1)	2.65(0.12)	0.2(0.1)	34	31	44	38
	BO_3 nonring	14.0(2.8)	2.53(0.12)	0.3(0.1)	16.8			
	BO_4 (1)	-0.3	0.45	0.6	47.4	69	56	62
	BO_4 (2)	-1.9	0.21	0.6	18.3			
^{17}O -Si-Zr glass	BO_3 ring	17.5(1)	2.65(0.12)	0.2(0.1)	27.6	35	56	46
	BO_3 nonring	14.0(2.8)	2.53(0.12)	0.3(0.1)	20.3			
	BO_4 (1)	-0.2	0.57	0.6	40.9	65	44	54
	BO_4 (2)	-2.0	0.10	0.6	11.2			

Under parentheses are given the standard deviation of the Gaussian distributions use for each parameters.

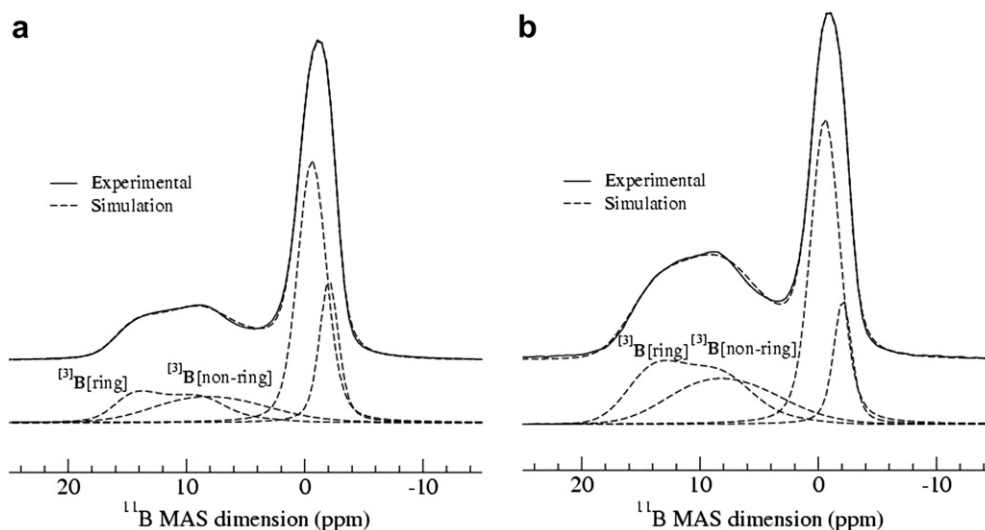


Fig. 3. Experimental (solid lines) and simulated (dashed lines) ^{11}B MAS spectra of the ^{17}O -Si (a) and ^{17}O -Si-Zr (b) glasses.

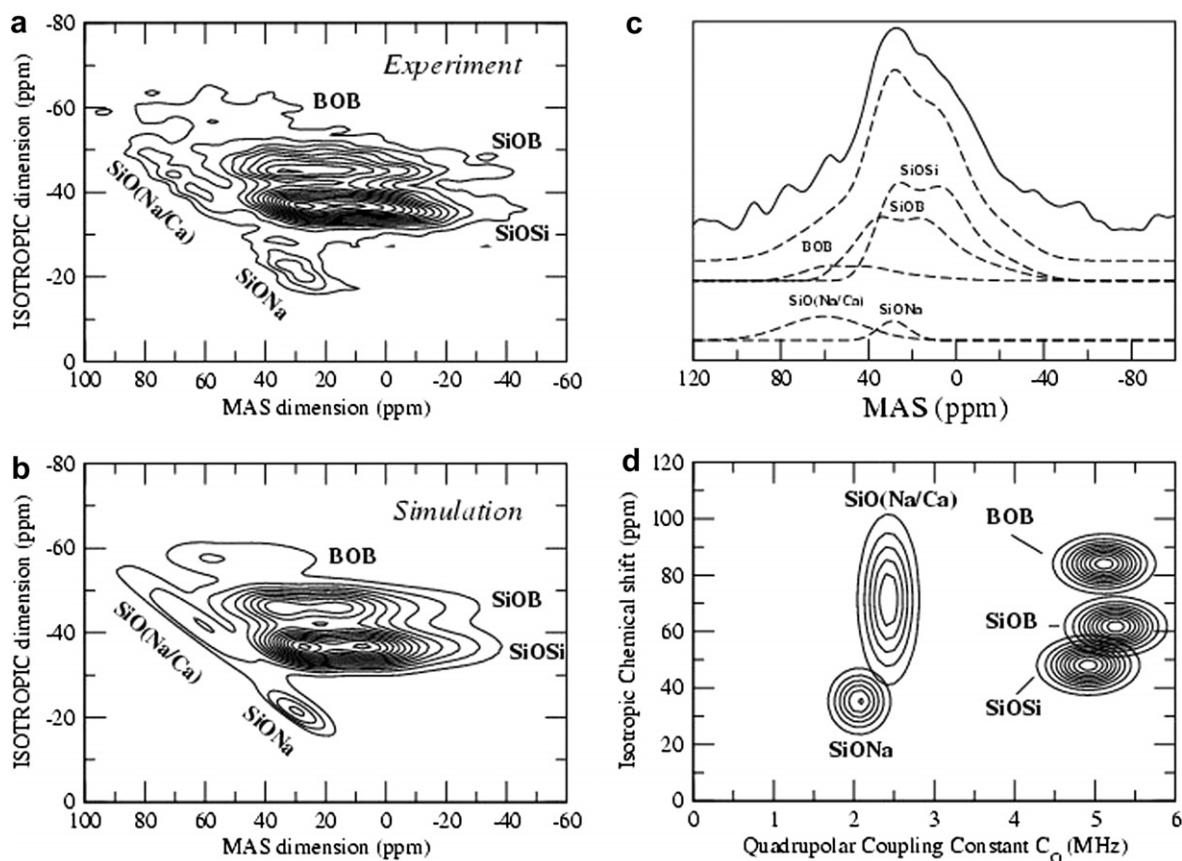


Fig. 4. (a) Experimental and (b) simulated ^{17}O 3QMAS spectrum of ^{17}O -Si Glass. (c) MAS projection of the experimental (solid line) and simulated (dashed line) 3QMAS spectra; each component are displayed (dashed lines). (d) Distribution of NMR parameters in the $(\delta_{\text{iso}}, C_Q)$ plane of each site.

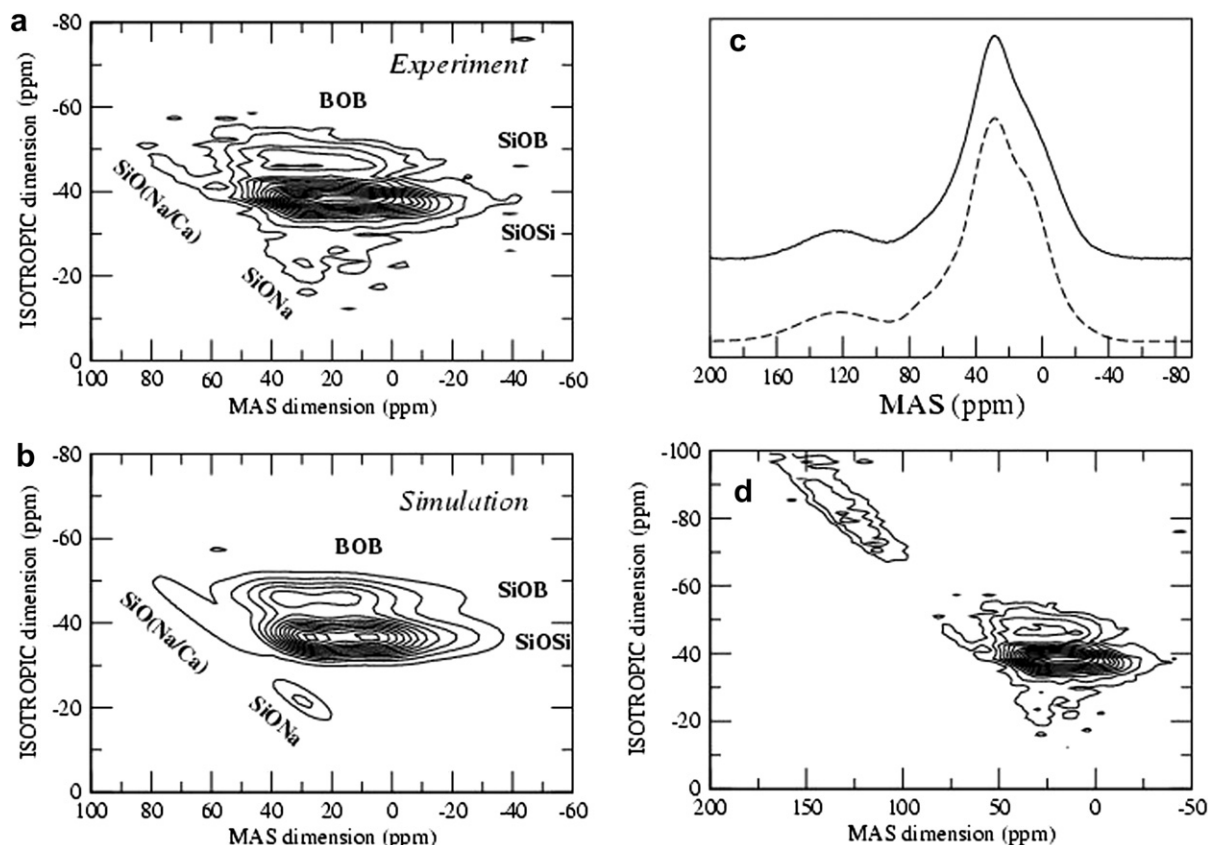
of the isotropic chemical shift and the quadrupolar coupling constant. All parameters are given in Table 7. The fact that such distributions can be related to topological distributions from known relationships between NMR parameters and local structural features (bond angles and distances) [31–33] is appealing for future prospects.

The case of ^{17}O -Si-Zr glass MQMAS spectrum is more complex. Indeed due to the folding back of the Si-O-Zr line in the isotropic

dimension (which has been corrected for displaying the spectrum in Fig. 5(d)), the Si-O-Zr region had to be treated separately. First, the experimental spectrum without the Si-O-Zr region (Fig. 5(a)) was fitted with the same NMR parameters as obtained from the ^{17}O -Si glass MQMAS experimental spectrum (Table 7) but with varying intensities. Subsequently, using the ^{17}O MAS spectrum (Fig. 5(c)), the relative population of the Si-O-Zr site has been calculated by fitting the Si-O-Zr line with a Gaussian ($\delta_{\text{iso}} = 123$ ppm

Table 7¹⁷O NMR parameters and quantification obtained from the *direct* simulation of the ¹⁷O–Si glass MQMAS spectrum

Site	C _Q (MHz)	σ _{C_Q} ^a (MHz)	η _Q	σ _{η_Q} ^b (MHz)	δ _{iso} (ppm)	σ _{δ_{iso}} ^c (ppm)	Intensity (%)
Si–O–Si	4.91	0.3	0.34	0.02	48.1	5.0	50
Si–O–B	5.24	0.3	0.45	0.02	61.9	5.0	36
B–O–B	5.07	0.3	0.46	0.02	84.3	5.0	8
Si–O–Na	2.12	0.2	1	0.58	35.2	6.0	1
Si–O–(Ca,Na)	2.60	0.2	0.89	0.58	70.5	17.2	5

^a σ_{C_Q} is constrained to be identical for Si–O–Si, Si–O–B and B–O–B sites, and for Si–O–Na and Si–O–(Ca,Na) sites.^b σ_{η_Q} is constrained to be identical for Si–O–Si, Si–O–B and B–O–B sites, and for Si–O–Na and Si–O–(Ca,Na) sites.^c σ_{δ_{iso}} is constrained to be identical for Si–O–Si, Si–O–B and B–O–B sites (precision of population is about 2%).**Fig. 5.** (a) Experimental ((d) with the Si–O–Zr region) and (b) simulated ¹⁷O 3QMAS spectrum of ¹⁷O–Si–Zr glass. (c) Experimental (solid line) and simulated (dashed line) MAS spectrum.

and σ_{δ_{iso}} = 24 ppm). Sites populations are given in Table 8. The broad distribution of the isotropic chemical shift for the Si–O–Zr site made it impossible to determine its quadrupolar parameters, but suggesting a value of the order of the NBO values or less. This is in agreement with a recent study pointing out the symmetric environment of Zr in a similar glass [34].

The NMR parameters of the bridging oxygen sites involving silicon and boron (Si–O–Si, Si–O–B and B–O–B) are similar to those previously reported in a sodium aluminoborosilicate glass [12]. Furthermore, the non-bridging sites Si–O–Na can be assigned unequivocally at around δ_{iso} = 35 ppm (C_Q = 2.1 MHz). However, the other non-bridging site (70 ppm, 2.6 MHz) can be related to a mixed peak Si–O–(Ca,Na) with a position depending on the Ca/Na ratio as previously observed in soda-lime silicate glasses [35]. These sites clearly show the predominance of a distribution according to the isotropic chemical shift interaction, as recently observed in aluminoborosilicate glasses containing potassium or calcium [10]. In SiO₂–ZrO₂ gels formed by the alkoxide process, the separation between the Zr–O–Zr, Si–O–Zr and Si–O–Si sites

Table 8¹⁷O quantification of the ¹⁷O–Si–Zr glass

Site	Intensity ^a (%)	Intensity ^b (%)
Si–O–Si	58	50
Si–O–B	32	28
B–O–B	5	5
Si–O–Na	1	1
Si–O–(Ca,Na)	4	3
Si–O–Zr	–	13

^a Intensities are obtained from the fit of the experimental ¹⁷O–Si–Zr MQMAS spectrum displayed in Fig. 5a, i.e. without the Si–O–Zr line, and using NMR parameters.^b Intensities including Si–O–Zr sites population, as obtained from the MAS spectrum of Fig. 5(c) (precision of population is about 2%).

was clearly visible on the ¹⁷O MAS spectra [36]. The Si–O–Zr site was situated near 150 ppm, similar to the value which is observed for the glass containing zirconium. To the best of our knowledge, this very well separated site from others is revealed for the first

time in a borosilicate glass using two-dimensional MQMAS NMR. The site position confirms the absence of other coordination for Zr (above the limit of detection) in glasses with potentially more charge compensators (Na^+ or Ca^{2+}) than $[\text{ZrO}_6]^{2-}$ groups.

Du and Stebbins [9,10] proposed two approaches for theoretically computing the proportions of various groups from the chemical composition of aluminoborosilicates. A random distribution of the groups in the glass structure is assumed or, conversely, some groups are considered as unable to form. We adapted these models to our compositions; the fact that bonds are prohibited between four-coordinate elements (except for silicon) was extended to bonds between zirconium and four- (except for silicon) or other six-coordinate groups. With these hypotheses the following groups are prohibited: $^4\text{B}-\text{O}-^4\text{B}$, $^6\text{Zr}-\text{O}-^6\text{Zr}$, and $^6\text{Zr}-\text{O}-^4\text{B}$. We assumed that all the zirconium is in octahedral configuration. The proportion of four-coordinate boron taken into account was the value measured by ^{11}B MAS NMR. The results are indicated in Table 9.

The random distribution model provides a better agreement, although the differences with the experimental data are still significant. The Si–O–B bonds are overestimated by the models in both glasses, whereas the Si–O–Si are underestimated, suggesting that the silicate and the borate network are not completely mixed. The higher experimental value of Si–O–Zr bonds could be in agreement with the lower proportion of B–O–B bonds if the presence of B–O–Zr bonds is considered; under these conditions, no distinction could be made in the ^{17}O MQMAS spectra between Si–O–Zr and B–O–Zr groups. The formation of B–O–Zr bonds leaves more silicon available for the formation of Si–O–Si sites and less boron available to form Si–O–B and B–O–B bonds. This could also account for the broad distribution of this site over the MQMAS spectrum, which could be related to the presence of both Si–O–Zr and B–O–Zr. In the same way, it has been suggested that adding aluminum to a borosilicate glass could lead to the formation of B–O–Al–O–Si units [37].

The non-bridging sites are largely underestimated by the calculation—in particular for glass containing no Zr (about 5% of NBO excess). NBOs arise mainly from Si–O–(Ca,Na) site (three to five times more numerous than sodium only bonded NBOs). The excess of a few percent of NBO fraction has already been observed experimentally in aluminosilicate [38] and borosilicate [39] glass compositions and attributed to the presence of oxygen triclusters compensating the NBOs. These observations were confirmed by *ab initio* molecular dynamics calculations [40,41]. The quadrupolar coupling constant that could be assigned to this type of site (about 2.3–2.6 MHz) would make it relatively difficult to observe here due to the proximity with non-bridging sites. Non-bridging sites could also be found with tricoordinate boron (BO_3) $_2\text{O}(\text{SiO}_3)$ and be indistinguishable on the MQMAS spectrum with bridging oxygen bonded to silicon [42]. No such excess of NBOs was observed in the glass without boron and calcium ($24\text{Na}_2\text{O}-76\text{SiO}_2$, see Section 3.1).

3.3. Gel structure

The same glass compositions than those examined in the preceding section were synthesized without prior oxygen-17 enrichment and then leached in H_2^{17}O so that ^{17}O enrichment occurred by transfer during leaching, yielding data on the structure of the alteration layer only. For a better understanding of how the glass composition can affect its reactivity, four additional glasses with different Na/Si ratios were also leached. The term ‘series’ is used in the following to represent two glass samples both with the same Na content, but only one contains Zr. The leaching times varied from one glass series to another, but were always identical within each series. The leaching times were specified on the basis of a previous study in which compositions near those of series 1 in Table 2 (18% Na_2O) were tested [3]; after a few tens of days, following a rapid drop in the leach rate, glass alteration practically ceased. Under the present experimental conditions at a high SA/V ratio with

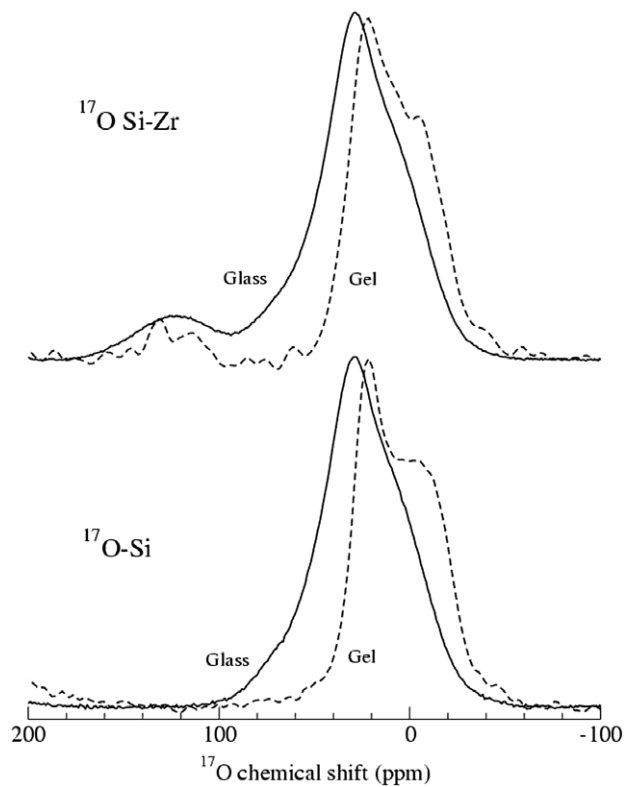


Fig. 6. ^{17}O MAS NMR spectra of pristine enriched oxygen-17 glasses (^{17}O -Si glass, ^{17}O -Si-Zr glass) and non enriched glasses altered in H_2^{17}O solution (Si gel 3, Si-Zr gel 3) Normalized to the same maximum peak height.

Table 9 Proportions of different sites computed from a random distribution model and a model prohibiting certain bonds (refer to text)

	Si–O–B	Si–O–Zr	Zr–O–B	Si–O–Si	B–O–B	Zr–O–Zr	NBO
^{17}O -Si							
Random mixing	43.2	–	–	46.3	10.1	–	0.5
$^4\text{B}-\text{O}-^4\text{B}$ avoidance	51.3	–	–	42.2	6.0	–	0.5
Experimental	36	–	–	50	8	–	6
^{17}O -Si-Zr							
Random mixing	36.2	7.2	3.0	43.9	7.5	0.3	2.0
$^4\text{B}/^6\text{Zr}-\text{O}-^4\text{B}/^6\text{Zr}$ avoidance	41.0	9.2	1.5	40.5	5.8	–	2.0
Experimental	28	13		50	5	0	4

unconstrained pH, these conditions were expected to be reached more rapidly. To ensure these conditions were reached, the glass specimens from series 1 were leached for six months, whereas the other series with higher Si content (lower chemical durability) were leached for two years. Table 2 indicates the molar compositions of the glass and the compositions of the alteration layers (determined from the elements released into solution). The retention factor is a measure of element retention in the alteration layer compared with boron (which is considered not to be retained at all).

Fig. 6 compares the ^{17}O MAS spectra of the pristine glass (Table 1) and of the resulting gels (series 3, Table 2). A decrease of the mean Si–O–Si peak position is observed related to the leaching of the Si–O–Na, Si–O–B and B–O–B sites in the gels. ^{17}O 3QMAS spectra were obtained for the altered glass samples (series 1) in which the glass containing Zr exhibited greater alteration (Fig. 7). The presence of a Si–O–Zr site in the gel is clearly visible very near

the site observed earlier in the pristine glass, although with a narrower distribution that is probably related to the greater symmetry around the Zr site in the gel. The narrower peaks, also observed for aluminum in the alteration layers of aluminosilicate glass compositions, may be attributable to the presence of protons near the element: when water is heated and evaporates, the initial peak width is usually restored [2]. The octahedral coordination of the $[\text{ZrO}_6]^{2-}$ groups thus remains unchanged during leaching but the nature of the charge compensator may be modified as it is the case for $[\text{AlO}_4]^-$ groups [2]. Moreover, a narrow site (also visible in the MAS spectra) was observed near the NBO sites of the glass without Zr. It could be attributable to surface Ca precipitates (the altered layer no longer contains any Na). The incorporated Ca in the gel structure should be positioned as Zr charge compensator.

The MAS spectra of all the gels are shown in Fig. 8. In gels containing Zr the proportion of Si–O–Zr to Si–O–Si sites was smallest for the glass with the lowest alkali concentration (series 3). The Zr

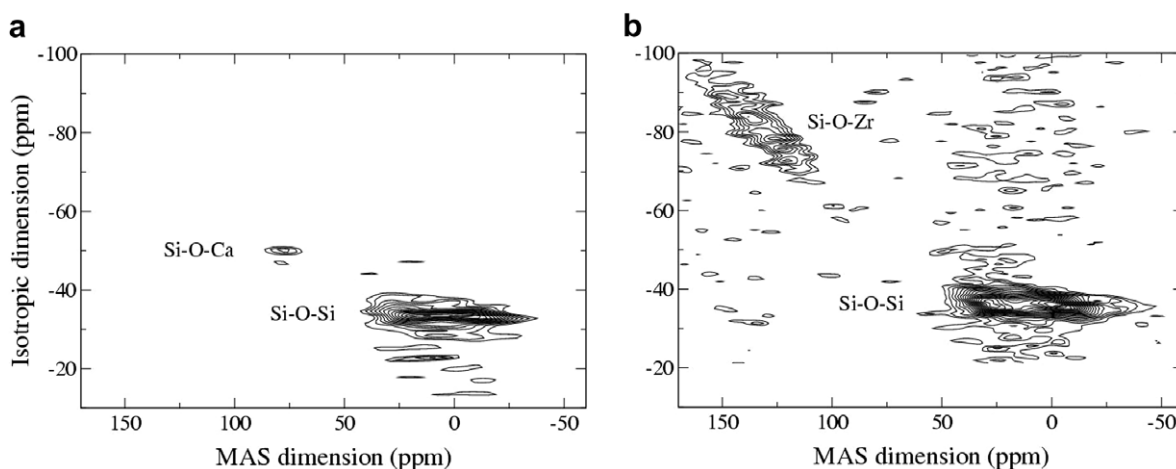


Fig. 7. ^{17}O 3QMAS spectra of Si gel 1 (a) and Si-Zr gel 1 (b).

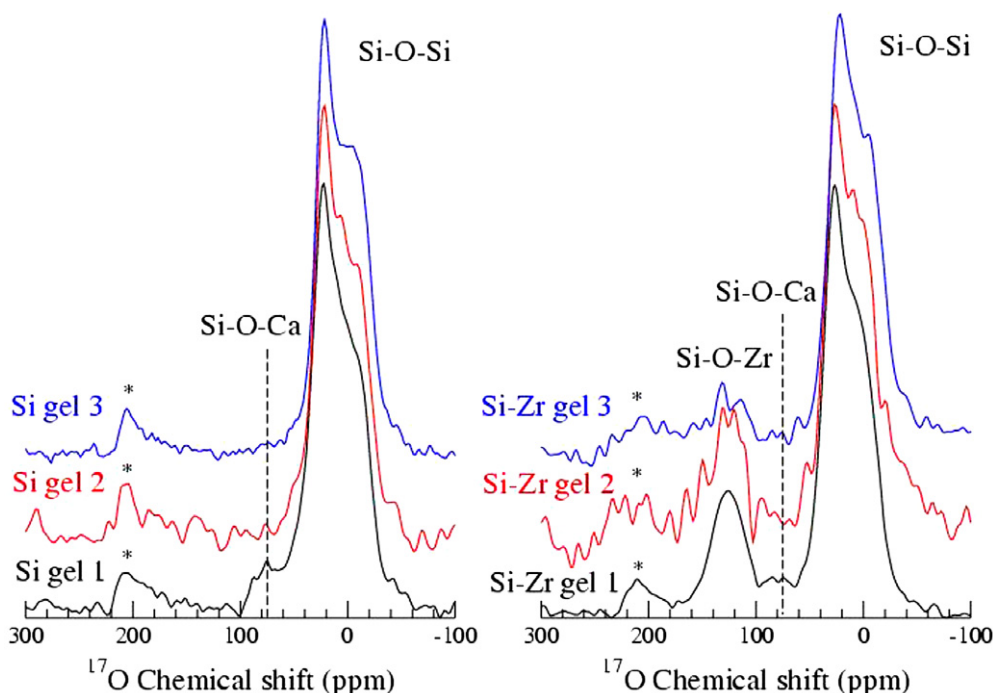


Fig. 8. ^{17}O MAS NMR spectra of altered glasses. Spectra are normalized to the same Si–O–Si maximum peak height.

sites are probably less accessible due to a more polymerized network. Conversely, in the least polymerized glass (the richest in Na), transfers were facilitated and the gel conserved only the charges (Na^+ and Ca^{2+}) necessary to compensate the $[\text{ZrO}_6]^{2-}$ groups (Table 2). In the other cases a slight excess of charges can form NBOs, although in quantities too small to be clearly identified in the NMR spectra. In all these gels Ca was found in larger amounts than Na, whereas the glass without Zr retained only Ca. This could imply that part of the Na is retained because of its Zr charge-compensating role, although most of the compensation is ensured by calcium in gels. When Na is only a network modifier (in the absence of Zr), it is entirely released into solution.

4. Conclusion

A method of quantitative analysis of MQMAS spectra have been presented, by a direct fit of the two-dimensional MQMAS spectrum which provides the resolution of all the structural groups forming the glass network. It was shown that ^{17}O 3QMAS is a powerful method for discriminating between the contributions of different structural groups and for quantitatively estimating their relative proportions. Additional data for the present analysis were also obtained from the quantitative deconvolution of the ^{11}B MAS NMR spectra, with the help of the MQMAS data. Alteration layers arising from glass leaching were also probed by oxygen-17 supplied by the leaching solution. Sixfolded zirconium is preferentially compensated rather than the tetrahedral boron and calcium only partially compensate the tetrahedral boron. Although most of the zirconium is inserted in the silicate network forming Si–O–Zr bonds, B–O–Zr bonds could also be present. Excess non-bridging oxygen was clearly identified in the glasses, and could be related to the presence of oxygen triclusters. During leaching, calcium clearly remains in the alteration gel, either stabilized near non-bridging oxygen or as a zirconium charge compensator, as zirconium conserves the same coordination number in the glass. This quantitative approach applied to ^{17}O MQMAS spectra have demonstrated its potential for investigating the structure of increasingly complex glass and gel compositions.

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